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Rethinking Career Sustainability Through the Lens of AI Affordance: The Exploratory Role of Knowledge Sharing

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Abstract

Artificial intelligence (AI), a transformative force, has revolutionised various aspects of human life and business operations. This has led to a drastic mutation of the career landscape, embedded with vast opportunities as well as challenges, particularly concerning career sustainability (CS). Despite myriad studies on CS, the paradoxical interplay of AI and CS remains underexplored, particularly for expatriates (expats). To address the aforementioned gap, our study incorporates an affordance perspective (AFP), positioning AI as an object and CS as a user context. Specifically, this study investigates whether AI facilitates the orchestration of an enhanced sustainable career within the boundary conditions of knowledge sharing (KS), encompassing both tacit and explicit knowledge pertinent to AI, cultivated through managerial initiatives and employee-driven activities. The study conducted a quantitative survey among 490 expats working in AI-integrated environments in China. The results reveal a curvilinear (U-shaped) relationship between AI and CS, where AI affordance at a moderate level enhances career adaptability and skill development. However, digital affordances become complex beyond a certain threshold, creating several career concerns, such as job insecurity and role ambiguity. Furthermore, the moderating effect of tacit and explicit KS mitigates numerous career disruptions while fostering long-term career growth. The study framed AI as both a tool and a collaborator that illuminates the importance of AI–human intelligence (AI–HI) synergy and knowledge augmentation in navigating digital transitions. Moreover, implications for international career development and human-oriented digital transformation are also discussed.

Keywords: artificial intelligence; affordance perspective; tacit knowledge sharing; explicit knowledge sharing; AI–HI collaboration; career sustainability

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1. Introduction

Artificial intelligence (AI), as the most influential technology and core of the Fourth Industrial Revolution [1], has intruded into almost every aspect of human life. AI, with its advanced capabilities, leverages human intelligence to comprehend various work and life scenarios, make informed decisions, and perform possible actions [2]. The digital transformation is becoming more of an ecosystem, a societal challenge, and a necessity, rather than just an organisational issue, in this contemporary business era [3,4]. This revolutionary drift has posed a drastic mutation of the business model landscape [5,6] and workplace behaviour [7], raising alarming concerns about a sustainable career [8], especially for expatriates (expats), because of several contextual factors such as knowledge sharing

(KS) [9], AI–human intelligence (AI–HI) collaboration [8,10]. Sustainable career posits "the sequences of career experiences reflected through a variety of patterns of continuity over time, thereby crossing several social spaces, characterised by individual agency, herewith providing meaning to the individual" [11]. Although the extant career literature has focused on orchestrating sustainable careers for employees [9,12,13], hitherto, scarce evidence is available on resolving the paradigm between AI technology and sustainable career, where the future of job reports by the World Economic Forum 2018 predicted 75 million employees lay-off due to an increase in new technology penetration and expected a cumulative damage of 85 million jobs due to the upcoming wave of robotics [14]. Notwithstanding, AI has been enormously integrated to recreate value at different levels among different entities [4], more specifically into human resource management (HRM) to induce value for employees [15] and facilitate employees' CS [8,16]; however, it has also posed threats to employees involved in a quest to preserve their jobs [17]. Therefore, an in-depth analysis is still required to clarify CS through an unorthodox lens of AI.

While the efficient orchestration of AI into organisational systems lies in accepting the AI affordances associated with the technology [5], as in the case of AI–HI synergy [10], existing AI research has highlighted the crucial role of AI affordances in adopting cutting-edge technologies, characterising it as a relationship between an actor and an object [5,10]. Affordance is "an action potential linked with an object that entails a specific user context" [18]. Considering the affordance notion, AI is an object, employees are the user context, and CS is the desired objective. Orchestrating a sustainable career through AI requires complex, dynamic capabilities that must be accomplished with sufficient skills and knowledge while mitigating societal risks, such as job insecurity [4,6]. Therefore, based on the above arguments, conducting this study through the theoretical lens of AFP appears to be a plausible approach. Furthermore, past studies have also embarked on developing and sharing new knowledge to harmonise AI's dual impact by fostering innovation through AI–HI collaboration, while enhancing career development [17] and organisational performance [5,19]. With the rapid digital transformation, AI is becoming a more significant augmentation force rather than an automation tool, reshaping work, knowledge, decision-making, and career transitions [20,21]. Subsequently, AI–HI interplay becomes central to career resilience within knowledge-intensive roles. Moreover, managerial initiatives (e.g., training programmes, mentorship) and employee-driven activities (e.g., peer learning, collaborative projects) can foster the AI-related tacit (e.g., problem-solving techniques, innovative applications) and explicit (e.g., technical skills, documented procedures) knowledge in the workplace, thereby augmenting AI affordance and moderating its impacts on career sustainability (CS). Scrutinising the above convictions, this study thus develops the following research questions:

1. What is the relationship between AI use and expats CS?
2. How does knowledge, i.e., tacit and explicit, influence the relationship between AI and CS?

A career ecosystem consists of three integral actors, i.e., the individual, the organisation, and the country. Therefore, to contextualise the study, expats were chosen as individuals working in several AI-integrated environments in China, a focus warranted by several converging factors. With a surge in rapid economic and technological development, China is the world's largest labour market and is expected to become the largest single economy by 2030 [22]. Since 2015, China's remarkable policy support and established technology standards have significantly boosted digital development, stimulating multinational businesses and attracting expats to work in China [9]. The context has been further intensified by the COVID-19 pandemic, which acted as a catalyst for surging the wave of digitalisation [23]. Expats, who are considered highly vulnerable to uncertainties faced in

the host country [24], have encountered numerous challenges in navigating this digitally accelerating environment, especially in the context of a distinctive Chinese organisational culture [25,26]. Concurrently, there is a transformative shift in AI technologies from automation tools to augmentation forces, particularly in knowledge-intensive scenarios [21,27]. This shift creates a need to understand how expats navigate such transitions to achieve both organisational resilience and CS. With approximately 600,000 expats residing in China for work, the majority in Beijing, Shanghai, and Guangdong provinces [28], and with regions like Zhejiang and Guangdong experiencing robust growth in high-tech sectors [29], China presents a unique and significant setting. Therefore, this study investigates AI affordances that must be embraced and the considerable knowledge augmentation required for effective AI–HI collaboration in this context.

This study's significant contribution lies in advancing prior research on AI and carers in three specific ways. First, the study demonstrates an inverted U-shaped relationship between AI affordances and CS, rather than assuming linear effects, which clarifies inconsistent findings in prior studies that alternately focused on AI-enabling opportunities or career threats [8,17]. Secondly, the study enriches theoretical understanding by distinguishing between tacit and explicit KS as conceptually distinct boundary conditions that shape the existence of AI affordances at different intensity levels, thus extending the existing AI–HI interaction and knowledge management research that conventionally treats KS as a unitary construct. Third, by focusing on expats in AI-integrated workplaces, the study extends sustainable career research to a vast, vulnerable, and underexplored workforce group that operates in digitally transformed economies and highly reacts to recent calls to contextualise CS across diverse employment settings [30].

2. Theoretical Foundation

2.1. Career Sustainability (CS)

As described earlier, CS posits a "sequence of career experiences reflected through a variety of patterns of continuity over time, thereby crossing several social spaces, characterised by individual agency, herewith providing meaning to the individual" [11]. This domain has captured much attention by academics in the past and has been proven to associate with several organisational and employee traits such as the following leadership styles: inclusive [31], paradoxical [32], career construction [33,34], career competence [35,36], career crafting [37], innovative behaviour [9], proactive [34,38], and work–leisure [39]. Amid the increase in globalisation [9,35,36], an unpredictable economic environment [40], global crises [31], and business digitisation [17,41], sustainable careers have emerged as a significant challenge for individuals and organisations to adapt to the rapidly changing competitive landscape. Considering the dynamic nature of CS, previous studies have proposed multidisciplinary scales to validate its characteristics. De Vos et al. [30] suggest three indicators, i.e., health, happiness, and productivity; Chin, Jawahar, et al. [42] streamlined it into four dimensions, i.e., flexible, renewable, resourceful, and integrative. Notwithstanding, the prior career literature has deeply investigated CS; however, the field is still in its infancy [12,30,42,43], more specifically, concerning business model innovation [5], AI adoption [17], and in the expats context [35]. Expats encounter unprecedented levels of career challenges [35,44], induced by several contextual factors such as cultural [35,45], knowledge management [46,47], and technological or digitalisation [36,48]. Owing to this fact, the literature on locals' CS seems vague in the context of expats, which calls for further investigation. Meanwhile, a study by Sobaih & Elnasr [49] identified different perceptions of local and foreign workers regarding their careers, including the nature of work, career development, commitments, and relationships with coworkers and managers. Hence,

concerning increased digitalisation and rapid economic growth, expats' CS remains a topic of debate and warrants further exploration.

2.2. AI and CS

AI is "the activity of developing machines with the ability to perform tasks that would normally require human intelligence, and intelligence refers to the quality that enables an entity to function appropriately and with foresight in its environment" [50]. AI, as a disruptive technology and part of the Fourth Industrial Revolution [51], holds significant potential to enhance the labour market and career ecosystem [52,53]. The pervasive penetration of AI into the business landscape has necessitated an urgent revision of employees' skills and knowledge [7,8,12,17], placing immense pressure on employees to adapt to dynamic changes and maintain their jobs [7,54]. While revamping the future of work, AI has brought numerous challenges for sustainable HRM, including managing the ethical risks and adverse human impacts of digitalisation, such as workforce displacement and the erosion of well-being [55,56]. Echoing this thought, numerous scholars have emphasised the need to nurture the CS issue in this digital era, particularly with respect to innovative work behaviour [9,57], AI–HI collaboration [8], knowledge management [12,17], leadership [31,32], and career competence [36]. Workplace digitalisation induced substantial benefits that enhanced employee experience and engagement at work [54,58]. Employees' positive attitude toward AI could foster their career goals by improving their skills, engaging in innovative behaviour, and identifying new opportunities [59]. Trust in AI constitutes strong employee–AI collaboration, which helps shape sustainable careers [8]. AI exerts value on the employees, enhances organisational efficiency, and improves performance [60]. Nonetheless, a viable scope exists for investigating AI implementation and its development pertinent to AI–HI collaboration [8,61], aiming for a sustained career in this digital era. Although the existing literature has focused on AI-driven automations, recent research emphasises the value of AI–HI collaboration in augmenting decision-making and knowledge creation [20,21,62]. Additionally, CS for expats within this dynamic technological realm is a debatable scenario to explore [35], as expats often require specialised assistance in a cross-cultural environment to adapt to changes in the work environment. Past studies have emphasised the development of new knowledge and the knowledge-sharing process as a vital source in enhancing AI–HI synergy [8], which can better exploit the benefits and opportunities associated with technologies [63], thereby initiating sustainable career patterns. The interplay between KS dimensions, i.e., tacit and explicit, remains underexplored in AI-augmented workplaces [64]. Surpassing task automation, AI has revamped the way knowledge is created and applied in the workplace. AI affordances pertinent to AI use can augment decision making and learning within a collaborative AI–HI ecosystem [64,65]. Thus, this study aims to address the challenges of accepting AI affordances that transcend traditional boundaries between humans and machines, creating an augmentation that either contributes to or undermines sustainable career development.

2.3. Affordance Perspective (AFP)

The AFP was coined early by Norman [66], who described it as a relationship between an actor and an object linked to a set of objectives. The relational aspect between technology and humans has gained significant importance, particularly concerning AI's inherent complexity and its far-reaching influence when implementing AI systems for making organisational decisions [10]. Thus, affordance highlights, "To what extent can a person or an organisation use a technology to accomplish a specific goal?" [18]. Affordance is more precisely stated as the action potential (e.g., the opportunity or motivation for carrying out such an action) provided by an object that represents the connections or exchanges

between the object (e.g., cutting-edge technology with particular features) and an actor (e.g., person or business) [67]. The theory holds significant importance in various technological domains, including information systems [68,69], AI–HI synergy [10], AI [5], and innovation and entrepreneurship [67], while also highlighting the missionary role of embracing AI affordances in utilising technology to achieve the required objectives. In this study, AI affordances refer to action possibilities that are offered to employees in their work contexts through AI technologies, which are contextualised at the theoretical level. Empirically, all these affordances are operationalised through a validated AI-use scale, which reflects employees' perceived intensity of using AI at work. At the same time, AI–HI collaborations provided a broader contextual understanding of the study, rather than being modelled as a separate construct.

3. Hypotheses Development

3.1. Impact of AI on CS

A plethora of studies have addressed the recreational role of technology in fostering career development practices [5,8,9,36]. However, the findings were inconsistent in resolving the paradoxical relationship between AI and CS [17,52]. As a matter of fact, AI has been integrated into HRM ecosystems to create value for employees [15], while enhancing their working experience and engagement at the workplace [54]. Moreover, workplace digitalisation has favoured sustainable careers, making employees more innovative towards their work [5,9]. Kong et al. [8] emphasised building trust in AI through fruitful AI–HI collaboration that maximises the interest in a sustainable career. Subsequently, a healthy collaboration with AI adds value to employees, makes organisations efficient, and improves performance [52,53,61], implying a sustained career for employees. AI, with its practical and efficient characteristics, has dramatically revamped the whole career landscape [36,52].

Despite the enormous advantages, AI also poses substantial threats to individual careers and job stability [7,17,52,56]. This is particularly true, as the updated AI wave is disrupting labour markets more rapidly and on a vaster scale than previous technological shifts [70]. AI, on the one hand, can enhance efficiency, personalisation, and learning opportunities, but its excessive adoption can reduce human agency and increase dependence, ultimately leading to de-skilling [71]. The successful integration of AI highly depends on employee willingness and organisational leadership's efficacy in ethically endorsing AI [4,7]. With the remarkable advancement of AI, fears about robots replacing humans have gained significant traction [17]. Ample studies have signified job stability as a crucial challenge to CS in this contemporary era, characterised by rapid technological advancements [72,73]. Moreover, implementing AI in HRM practices introduces a range of ethical complexities, including fairness, privacy, and bias [7,74].

Drawing upon the AFP, technological use can create both enabling and constraining action possibilities that emerge during the interaction between human and machine [75]. AI use at a lower level can enhance CS by expanding employee functional capabilities, adaptivity capacity, and learning opportunities, which support resource renewal and long-term employability [30]. However, AI use intensifies beyond an optimal level, and excessive reliance on AI may reduce individual autonomy, perceived career control, and skill development, thereby leading to capability erosion and reducing CS. This paradigm between capability enhancement and capability erosion suggests a non-linear process, resulting in an inverted U-shaped relationship between AI use and CS. Thus, to address RQ 1, this study hypothesises the following:

H1: *The relationship between artificial intelligence (AI) use and CS is inverted U-shaped.*

3.2. Artificial Intelligence and Knowledge Sharing (Tacit and Explicit)

A vast amount of the literature stressed resolving the paradoxical relationship between AI and KS [5,76–78], which has revitalised and reshaped various concepts such as employee productivity [79], HRM practices [76], employee innovative behaviour [5,9], organisational innovative behaviour [3], business performance [19], organisational creativity [41], and firms' decision-making [77]. The crux of the above practices had a significant impact on employee career prospects [17,24,32]. New and existing knowledge of technology rigorously builds up employees' intellectual capital [9], encouraging them towards a prospective career through innovative thinking capability. Employees who are aware of their career development should continually acquire and apply new knowledge at work [80]. Moreover, the unique knowledge of employees makes them competent and worthy in the workplace [81]. A recent finding on overcoming the threats of robots replacing humans highlights the paramount importance of unique knowledge through efficient knowledge management, which maintains human supremacy over machines by leveraging technology as a source of knowledge and skill development [17].

Communication, sharing, transfer, or coordination of knowledge or expertise within the organisation occurs via a channel characterised as KS [82]. Based on a taxonomic division of organisational knowledge, numerous studies categorised unique knowledge into explicit and tacit [61,83]. Organisational knowledge is grounded in the interplay of tacit and explicit knowledge, thereby facilitating the creation of new knowledge to manage innovations, performance, and strategies [61,76,84]. Tacit knowledge, such as informal mentoring or experiential insight, is gained through experience and unconsciously integrated into one's identity [85]. Conversely, explicit knowledge, such as documented procedures or databases, is the codified knowledge in the form of documents or reports, preserved and shared through any technological means within the organisation [86]. To leverage an advanced knowledge-sharing exchange and effective AI–HI interaction, firms adapt AI-enhanced HRM tools to exchange explicit and tacit knowledge [76]. However, the post-pandemic era has adversely affected the applicability of tacit KS due to increased remote working trends, diminishing the formation of new knowledge exchanges [78] and posing threats to employees' technological intelligibility and career growth. Moreover, the KS mechanism among locals and expats appears challenging due to the tacit nature, complexity, and specificity, which requires rigid organisational support to harness expats' unique knowledge [63]. Expats are considered a bridge in leveraging and sharing unique cross-cultural knowledge at the workplace, thereby increasing knowledge intelligence [87]. Hence, an exploratory study on KS among expats to develop AI-integrated sustainable careers possesses great significance across borders.

Comprehending the study arguments under the theoretical lens of AFP, this study assumes that perceived technological affordances have an impact on outcomes only when they are actively actualised through cognitive and social practices [88]. Explicit KS can facilitate the codification and scalable transfer of AI-related insights, which enhances initial performance and learning benefit conveyed with a moderate level of AI use. Contrary to this, tacit KS emphasises experiential learning, adaptive interpretations, and contextual sense making, which buffer the negative consequences associated with extensive AI reliance at the highest intensity levels and sustaining career-related resources over time [86]. Thus, to address RQ 2, this study hypothesises the following:

H2a: *Tacit KS moderates the inverted U-shaped relationship between AI use and CS, such that the inverted U-shaped relationship is stronger when tacit knowledge is high rather than low.*

H2b: *Explicit KS moderates the inverted U-shaped relationship between AI use and CS, such that the inverted U-shaped relationship is stronger when explicit knowledge sharing is high rather than low.*

4. Methods

4.1. Sample and Data Collection

An online quantitative survey was administered to expats working across diverse industries in China to assess the applicability of this study's theoretical model. Participants were recruited through professional expat networks, organisational contacts, and online platforms commonly used by expats in China. The inclusion criteria ensured the relevance of participants, who were mainly employed in AI-integrated work environments, as reported by the participants themselves. The survey was distributed in key metropolitan areas, namely Beijing, Shanghai, Zhejiang, and Guangdong Province, which are considered to be the home to the largest concentrations of the target demographics [28]. The regions, particularly Zhejiang and Guangdong, have recently experienced the most robust population inflow, with subsequent growth in sectors such as finance, the internet, and high-tech industries [29]. This targeted approach facilitated the recruitment of a substantial sample from the estimated population of approximately 600,000 expats currently residing in China, thereby establishing these regions as suitable and relevant contexts for this study. Additionally, China has been identified as one of the most challenging in terms of careers for expats to live and work [26], primarily due to Chinese organisational behaviour [25], highly affirming the study context. The survey comprised two sections, demographics and study variables, with English chosen as a communication medium that caters to expat participants. Data were collected at two different time points. At Time 1, AI use and control variables were measured, and at Time 2, CS and KS variables were collected, which helped reduce the risk of common method variance [89]. During the first wave of the survey (T1), the questionnaires were sent to 645 expats willing to contribute to this anonymous survey and report their demographic profiles, AI adoption, and CS data. A total of 550 completed responses were received. This study gathered explicit and tacit KS information during the second wave survey (four weeks later). Ultimately, this study received 490 valid responses, yielding an effective response rate of 75.96%. The demographics of the response sample include a majority (58.8%) of males, with a majority 60.2% holding a bachelor's degree. The average age is 29.60 years (SD = 8.23).

4.2. Study Measures

All survey items were measured using a five-point Likert scale, i.e., 1 = strongly disagree, 5 = strongly agree.

4.2.1. Career Sustainability (CS)

Twelve items for CS were adopted from the study by Chin, Jawahar et al. One sample item was "My career enables me to have a good standard of living" ($\alpha = 0.95$).

4.2.2. AI Use

This study adapted 4 items from Kai Chi Yam et al. [90] for AI use. One sample item was "Today, many of the decision-making activities of this job are automated or assisted by AI technology" ($\alpha = 0.96$). The scale captures the perceived intensity of AI use rather than breadth or dependence.

4.2.3. Explicit and Tacit Knowledge Sharing

Moreover, tacit and explicit KS were adapted from Choi et al. [83]. Tacit KS was measured using a 4-item scale. One sample item was "Knowledge about AI tools is acquired through one-to-one mentoring in my company" (α = 0.96). Explicit KS was measured using a 4-item scale. One sample item was "Knowledge about AI tools (including know-how, technical skills, and problem-solving methods) is well codified within my company" (α = 0.95).

Further details are provided in Appendix A.

4.2.4. Control Variables

Additionally, this study incorporated age, gender, and education as control variables, following earlier studies [9]. Gender was coded as 1 for males and 2 for females, and age was measured in years. Education captured respondents' higher level of formal education achieved and was coded as 1 = below bachelor's degree, 2 = bachelor's degree, 3 = master's degree, and 4 = PhD or above. These variables were included to measure the potential demographic influence on CS.

4.3. Construct Validity

Initially, the study performed a series of confirmatory factor analyses (CFAs) to determine the relationships among the following studied constructs: AI, explicit KS, tacit KS, and CS, using Mplus 8.7. As delineated in Table 1, the study's four-factor model significantly fit the data well as follows: χ^2 (246) = 941.56, (CFI) = 0.95, (TLI) = 0.94, (RMSEA) = 0.07, and (SRMR) = 0.04, which was better than all other tested models. The complete details are prescribed in Table 1.

Table 1. Comparison of structural models.

Models	Factors	χ^2	d.f.	χ^2/d.f.	CFI	TLI	RMSEA	SRMR
Model 1	Four factors: AI, Explicit KS, Tacit KS, and CS	941.56	246	3.82	0.95	0.94	0.07	0.04
Model 2	Three factors: Explicit KS and Tacit KS combined into one factor	3121.19	249	12.53	0.78	0.76	0.15	0.10
Model 3	Two factors: Explicit KS, Tacit KS, and AI combined into one factor	5351.47	251	21.32	0.61	0.58	0.20	0.20
Model 4	One factor: All variables combined into one factor	6748.58	252	26.78	0.51	0.46	0.22	0.16

Note: AI: artificial intelligence; Explicit KS: explicit knowledge sharing; Tacit KS: tacit knowledge sharing; CS: career sustainability; CFI = comparative fit index; TLI = Tucker–Lewis index; RMSEA = root mean square error of approximation; SRMR = standardised root mean square residual.

4.4. Assessment of Common Method Variance (CMV)

To mitigate the threat of common method variance (CMV), we gathered data on the predictors and outcome variable at different time points with a four-week interval to confirm that the answers in the first wave were less salient [89]. The study specifically deployed statistical techniques to avoid the threat of CMV. First, the study employed Harman's single-factor test, a widely applicable method in the management literature. The results revealed that the total variance exhibited by the single-factor method was 0.40, which is under the threshold limit of 0.50. Second, the study incorporates the unmeasured latent factor method approach to detect the CMV issues. We loaded a common factor into the CFA, whose estimates indicate the hypothetical framework that gives rise to the new

CMV model. The results confirmed that the new CMV model was not significantly better than the CFA four-factor model ($\Delta\chi^2$ (30) = 71.57, $p < 0.00$).

4.5. Convergent and Discriminant Validity

The convergent validity of the variables was measured using the average variance extracted (AVE). All the AVE values were higher than the 0.5 threshold. Hence, both convergent and discriminant validities were identified in Table 2.

Table 2. Convergent and discriminant validity.

Constructs	AVE	MSV	1	2	3	4
			AI	EK	TK	CS
1. CS Career Sustainability	0.669	0.433	0.818			
2. EK Explicit Knowledge	0.843	0.433	0.658 ***	0.918		
3. AI Artificial Intelligence	0.858	0.178	0.422 ***	0.292 ***	0.926	
4. TC Tacit Knowledge	0.875	0.195	0.283 ***	0.442 ***	0.405 ***	0.935

Notes: (along diagonal are square roots of AVE, off-diagonals are inter-construct correlations) significance of correlations: *** $p < 0.001$.

5. Results

Descriptive Statistics

The means, standard deviations, and correlations of the measured variables are given in Table 3.

Table 3. Means, standard deviations, and correlations.

Variables	M	SD	Skewness	Kurtosis	1	2	3	4	5	6	7
1. Gender	1.41	0.49	0.35	−1.88							
2. Age	29.60	8.23	1.19	0.88	0.00						
3. Education	2.38	0.79	0.72	−0.07	0.03	−0.09					
4. AI	3.81	1.15	−0.79	−0.14	0.06	−0.07	0.00	(0.96)			
5. Explicit KS	3.37	1.41	−0.56	−1.37	0.04	−0.01	0.01	0.28 **	(0.95)		
6. Tacit KS	3.62	1.27	−0.66	−0.60	0.06	0.01	−0.05	0.38 **	0.42 **	(0.96)	
7. CS	3.66	1.02	−0.72	−0.95	0.06	0.00	0.07	0.40 **	0.62 **	0.27 **	(0.95)

Note: $N = 490$; internal reliabilities (Cronbach alpha coefficients are specified along the diagonal parenthesis), ** $p < 0.01$. Gender was coded as 1 = male, 2 = female. Education represents the highest qualification achieved (1 = below bachelor's, 2 = bachelor's, 3 = master's, 4 = PhD or above); AI: artificial intelligence; Explicit KS: explicit knowledge sharing; Tacit KS: tacit knowledge sharing; CS: career sustainability.

The study performed the hierarchical regression analysis to test the relationships among conceptual constructs, and the results are delineated in Table 4. Model 1 and Model 2 exhibit the main effect and quadratic terms for AI, respectively. The results indicate that the coefficients for the main impact of AI use (b = 0.34, SE = 0.05, $p < 0.001$) and the quadratic term report the negative coefficients of AI (b = −0.16, SE = 0.04, $p < 0.001$) and are statistically significant, supporting Hypothesis 1 (AI has an inverted U-shaped effect on CS). Hypotheses 2a and 2b propose that explicit KS and tacit KS moderate the inverted U-shaped relationship between AI and CS. To test the proposed associations, this study added the interaction effects of the squared term of AI use and explicit KS in Model 3 and that of AI use and tacit KS in Model 4. The results of Model 3 reveal that the interaction term between the quadratic effect of AI use and CS is statistically significant (b = −0.16, SE = 0.03, and $p < 0.001$), supporting Hypothesis 2a. Likewise, the results of Model 4

show that the interaction term between the quadratic effect of AI use and CS is statistically significant (b = −0.07, SE = 0.03, $p < 0.05$), supporting Hypothesis 2b.

Table 4. Regression analysis.

	Model 1		Model 2		Model 3		Model 4	
Variables	**Coeff.**	**SE**	**Coeff.**	**SE**	**Coeff.**	**SE**	**Coeff.**	**SE**
Gender	0.06	0.12	0.07	0.12	−0.01	0.09	0.02	0.11
Age	0.00	0.00	0.00	0.00	−0.00	0.00	0.00	0.00
Education	0.03	0.07	0.02	0.07	−0.06	0.06	0.04	0.07
AI	0.34 ***	0.05	1.42 ***	0.28	−2.98 ***	0.67	−0.41	0.70
AI^2			−0.16 ***	0.04	0.40 **	0.10	0.02	0.10
AI * Explcit KS					1.20 ***	0.19		
AI^2 * Explicit KS					−0.016 ***	0.03		
AI * Tacit KS							0.60 **	0.20
AI^2 * Tacit KS							−0.07 *	0.03
R^2_{total}	0.08 **	0.02	0.10 ***	0.02	0.45 ***	0.03	0.25 ***	0.03
ΔR^2	0.08 **		0.02 *		0.35 **		0.15 **	

Note(s): $N = 490$; *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Explicit KS: explicit knowledge sharing. Tacit KS: tacit knowledge sharing.

For further clarification, the study followed the guidelines of earlier studies that have explored the curvilinear relationship [91,92]. This study plotted the two diagrams to further demonstrate the moderating effects of explicit KS and tacit KS, at low (−1 SD) and (+1 SD) in the relationship between AI use and CS, as shown in Figures 1 and 2. When explicit KS was higher, we found a positive effect at low levels of AI use (effect = 0.40, SE = 0.16, and 95% CI = [0.15, 0.70]) but not significant at high levels of AI use (effect = 0.07, SE = 0.09, and 95% CI = [0.22, 0.07]). The slope of the inverted U-shape of explicit KS on CS becomes steeper, as reported in Figure 1. Similarly, when tacit KS was higher, we found a positive effect at low levels of AI use (effect = 0.45, SE = 0.18, and 95% CI = [0.20, 0.82]) but not significant at high levels of AI use (effect = 0.20, SE = 0.16, and 95% CI = [0.06, 0.12]). The slope of the inverted U-shape of tacit KS regarding CS becomes steeper, as shown in Figure 2.

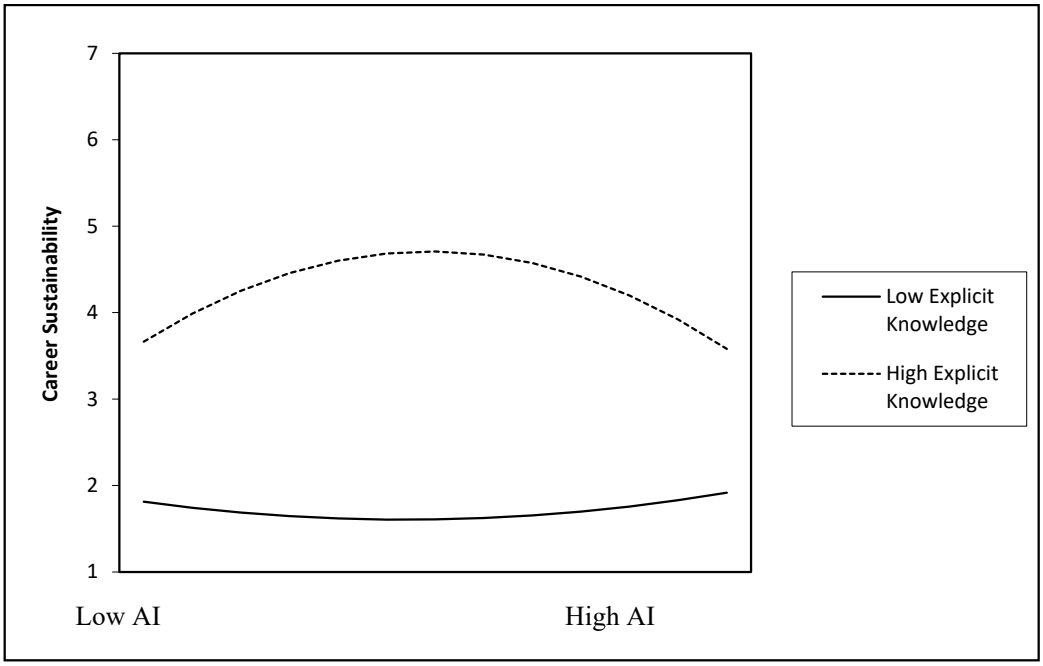

Figure 1. Positive moderating effect of explicit KS on the inverted U-shaped relationship between AI and CS.

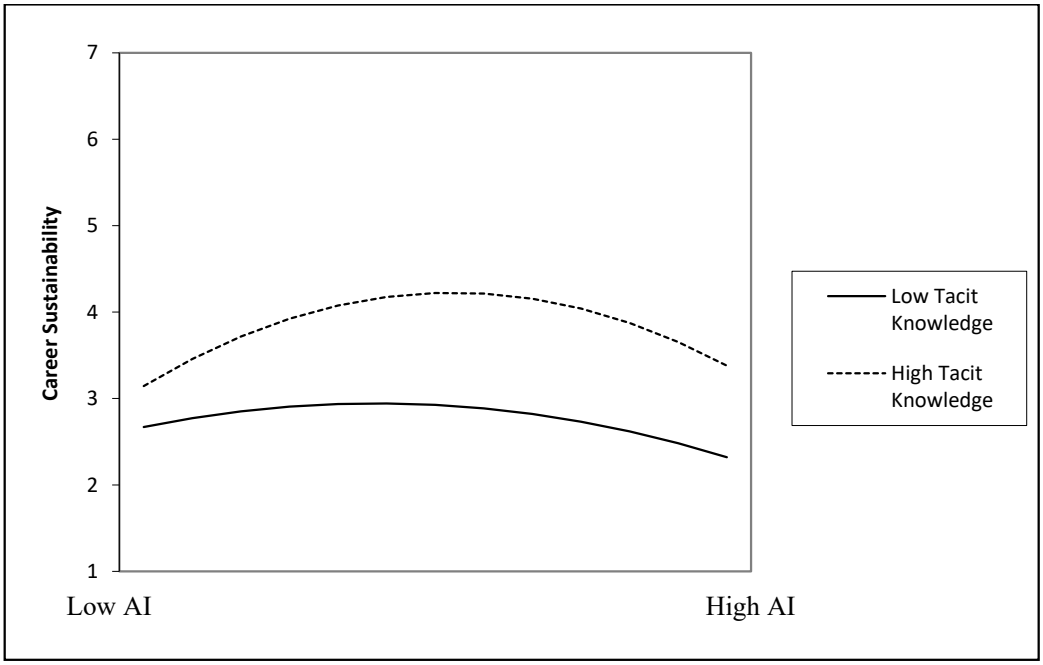

Figure 2. Positive moderating effect of tacit KS on the inverted U-shaped relationship between AI and CS.

6. Discussion

The study examines the intricate relationship between AI and CS through the theoretical lens of AFP, leading to the formulation of H1. Additionally, the study extends the arguments in H2a and H2b by delving into the imperative role of KS (i.e., tacit and explicit) as a moderator, facilitating the orchestration of AI-integrated CS. The empirical results robustly support all proposed hypotheses, revealing a curvilinear (inverted U-shaped) relationship between AI and CS, moderated by tacit and explicit KS but in distinct ways. Tacit KS attenuates the negative slope, whereas explicit KS strengthens the initial positive curve. Simply stated, AI affordances offer substantial career development opportunities, but their overreliance may lead to unsustainable outcomes. KS is found to be the key enabler in managing AI-driven transitions. Therefore, the study's findings adequately address the following formulated research questions: (1) How and to what extent do AI technologies facilitate the orchestration of a sustainable career?; (2) How do tacit and explicit knowledge influence the relationship between AI and CS? In summary, the current study enriches our understanding of expats' effective orchestration of AI-assisted sustainable careers through a consistent exchange of tacit and explicit knowledge in the workplace. The theoretical and practical implications are discussed below.

6.1. Theoretical Contributions

This study makes four distinct theoretical contributions to the literature on AI, CS, and knowledge management. First, it extends AFP to the domain of CS. Second, it advances AI and career research by demonstrating a nonlinear, inverted U-shaped relationship between AI use and CS. Third, the study findings differentiated the distinct moderation roles of tacit and explicit KS in AI-integrated work contexts. Fourth, it enriches the CS literature by focusing on expats as a vulnerable and underexplored workforce segment in digitally transformed economies.

6.1.1. Theoretical Extension of AFP

Our study makes a vital contribution to extending the affordance paradigm within the amalgamation of digital landscape and the career literature, clarifying the way to attain sustainable careers through AI technologies. The findings suggest a curvilinear relationship between AI use and the effective orchestration of advanced, sustainable career setups. AI affordances bear a dynamic and multifaceted characteristic in adopting and utilising cutting-edge technologies [5,67] that revamp the existing trajectory towards sustainable and productive careers. With its optimal affordance, AI initially provides extensive opportunities for skill development and career prospects [8,59]. However, a surge in AI use beyond an optimal level makes the AI affordances complex, posing threats to individuals' career concerns, such as job insecurity [17,52]. Ultimately, our findings build upon conventional understandings of CS by facilitating technology orientation into careers through the theoretical lens of AFP [10] and AI–HI synergy [8], thereby enhancing the applicability of AI-integrated sustainable careers in today's contemporary era [7]. According to AFP, these findings suggest that AI contributes to CS not merely through adoption, but also through how employees actively perceive, interpret, and actualise AI-enhanced action possibilities in work contexts.

6.1.2. Nonlinear AI-Career Logic

Given the increasing infiltration of AI into organisations and society [4,15], these study findings suggest incorporating AI efficiently for innovative career solutions by deeply exploiting the AI affordances between cutting-edge technologies and CS [8]. Additionally, this study addresses the scattered findings on the AI and CS paradox, with some studies revealing positive relationships [8] and others predicting a negative impact [93]. Moreover, the study findings also demonstrate this paradox by demonstrating a curvilinear mechanism between AI and CS.

6.1.3. Differential KS Roles

Furthermore, this study vastly enriches the literature on knowledge management by delving into healthy AI–HI synergy [61] and forming a sustainable career through efficient KS [8]. Through tacit and explicit KS, this study narrated valuable insights on nurturing the relationship between cutting-edge technologies and career practices towards AI-enhanced CS [17]. The existing literature has linked KS with technology, framing it as a positive force that enhances both human and machine collaboration [5,12] and workplace performance [19,83]. However, our study distinctively configured KS through tacit and explicit knowledge levels through AFP, favouring an enhanced interaction between an object (cutting-edge technology) and an actor (employee) to attain particular objectives (CS). Thus, tacit and explicit KS should be the foremost consideration when implementing AI in the workplace [76], for better AI–HI interaction [19].

The study findings also contribute to the ongoing debates on AI-driven knowledge augmentation. Explicit KS primarily strengthens the initial performance and learning benefits of AI use, whereas tacit LS buffers the negative career consequences that can emerge at higher levels of AI intensity. AI affordances, coupled with collaborative knowledge practices, may attenuate work displacement and enhance organisational resilience and inclusiveness, which are considered core to effective digital transformation [94].

6.1.4. Expats Sustainability Insight

This study extends the theoretical perspective on career research by foregrounding expats as a workforce disproportionately exposed to AI-driven career vulnerability. Simply stated, expats encounter amplified career vulnerabilities under high AI adoption, depicting

global inequities in digital transformation. This directs policymakers to address these discrepancies through an advanced AI upskilling initiative. Moreover, in alignment with open innovation paradigms, tacit KS, such as mentorship, enhances the co-creation of AI-augmented solutions. Organisations leveraging such practices can simultaneously improve employee productivity and well-being.

Lastly, these study findings also contribute to promoting diversity in the workplace, with a specific focus on the sustainable career dynamics of expats in a fast-moving digital economy such as China [22], which accommodates more than 600,000 expats in different cities [28]. Additionally, China is considered challenging in the career context for expats, with a high possibility of expats' failure [26]. Therefore, our findings identify the core environmental conditions that can foster a healthy synergy between cutting-edge technologies and expats in the workplace. Both tacit and explicit KS on realising and accepting the AI affordances can help expats build a sustainable career.

6.2. Practical Implications

Considering the nuanced understanding of the AI-integrated sustainable career paradox, this study offers paramount implications to practitioners, entrepreneurs, and businesses, including multinational setups. Organisations should develop AI training programmes to codify tacit knowledge, such as peer-learning platforms, thereby sustaining innovation while overcoming career disruptions. Revamped policies on facilitating cross-sector partnerships to promote and reskill the vulnerable workforce, such as expats and gig workers (independent, platform-based contractors engaged in short-term work), in economies with extensive AI adoption. For instance, an HR manager can design AI onboarding programmes that combine technical training with structured mentorship arrangements, enabling expats to acquire both explicit AI knowledge and tacit, context-specific insights. Similarly, organisations may employ AI–HI hybrid work systems, where AI supports routine tasks while employees focus on national and judgement-intensive work, which may reduce career threats. Additionally, policymakers in AI-enhanced economies should support reskilling initiatives aligned with national AI strategies to mitigate the unequal risks faced by expats.

Secondly, pertinent to AI affordances' foremost role in revamping the complex relationship between AI use and CS, employees and leaders should focus on understanding and embracing the affordances associated with cutting-edge technologies to effectively exploit the outcome benefits.

Thirdly, the unprecedented volatility in career prospects resulting from AI's significant impact on businesses poses a threat of machines replacing humans [52]. Thus, contemporary knowledge of AI affordances will allow firms and employees to use technology as a companion rather than a competitor or threat [17].

Lastly, the efficient implementation of cutting-edge technologies, such as AI, at the workplace corresponds to the CS of employees, which fosters innovative behaviour at the workplace [5], thereby enhancing organisational performance [19] and achieving a competitive advantage in the industry.

7. Conclusions

The rapid evolution of AI, equipped with cutting-edge analytical and logic-based tools, has revamped orthodox business ideologies, raising a paradoxical concern regarding expats' CS by creating both opportunities and threats. While AI offers significant advantages in enhancing employee efficiency and productivity, it raises critical concerns about job stability due to ethical complexities such as fairness, biasness, privacy, and the fear of machines replacing jobs. Another dilemma in embracing cutting-edge technologies is

the insufficiency of new knowledge and its dissemination among employees, which is essential for effective AI–HI interaction. The study also provides timely insights into AI–HI dynamics by demonstrating the interaction between AI use and KS in designing CS, while contributing to the broader discourse on digital transformation and knowledge augmentation. Future scholars are encouraged to investigate the evolution of AI–HI collaboration and how a human-centred, inclusive AI design can effectively balance equitable career development in contemporary digital society. Additionally, the local CS literature available seems vague when applied to the expats' context. Thus, this study clarifies the notions behind the effective orchestration of AI-integrated CS aligned under the AFP, positing the need to accept AI affordances and enhance KS, i.e., tacit and explicit, that enhance the AI–HI interaction and facilitate employees in using AI. By empirically demonstrating an inverted U-shaped relationship between AI use and CS, this study clarifies why AI adoption simultaneously creates opportunities and risks for expats in the long term. More importantly, the findings reveal that the explicit and tacit KSs play a distinct yet complementary roles in sustaining careers at different levels of AO intensity. These insights prove that AI is not a pure technological intervention, but a socio-technical system whose career implications depend largely on human-centred knowledge practices. Overall, the accumulated findings of this study contribute to advancing career research by integrating the AFP with knowledge management theory, while offering practical insights for organisations seeking to design inclusive, human-centric AI-enabled workplaces.

8. Limitations and Future Scope

Despite the ubiquitous implications of this study's unique findings, several contextual, methodological, and measurement-related limitations should also be noted for future research.

The data on expats working in AI-integrated environments from a single country, China, may present generalizability issues to other national and institutional contexts. Although China represents a highly relevant setting due to its rapid diffusion of AI, analysing data across multiple countries and institutional environments could enhance the external validity of the study's findings. Moreover, industry and cultural background variables were not included as focal moderators in the current model, as this study's primary focus was on uncovering generalizable AI affordance mechanisms rather than context-specific effects. Considering the dispersed nature of expats across several industries and nationalities, modelling such heterogeneity demands larger sub-samples to ensure statistical robustness. Thus, culture and industry are treated as boundary conditions rather than explanatory variables. Future scholars are encouraged to initiate qualitative and mixed-method research to explore how cultural norms, industry logics, and national AI policy environments influence KS and CS in AI-integrated workspaces, particularly in emerging economies.

The study relies solely on survey-based data, which were collected using a two-wave design; however, it may still be subject to perceptual bias. Additionally, the quantity approach enables testing of nonlinear and moderating effects, but it still provides deeper insight into how individuals interpret and experience AI-related career dynamics. Future scholars are encouraged to employ longitudinal or qualitative designs to offer richer, process-oriented insights into how AI–HI interaction evolves and influences CS.

Finally, the data constraints and limited availability of validated measurement scales from emerging AI-related constructs may impose challenges for construct refinement. This study carefully adapted the established scales where possible. However, future research is still encouraged to develop further and validate context-sensitive measures of AI use, affordance actualisation, and KS mechanisms to enhance measurement precision, reliability, and validity.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study. Participation was voluntary, responses were anonymous, and no identifying information was collected.

Data Availability Statement: The data presented in this study are available on reasonable request from the corresponding author. The data are not publicly available due to ethical restrictions and the confidentiality assurances provided to participants.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Table A1. Study measures.

Scale	Items		Mean	Standard Deviation	Factor Loading	Cronbach Alpha	Composite Reliability (CR)	VIF
AI Adoption (Yam et al., 2023, [90])	1.	Today, many of the decision-making activities of this job are automated or assisted by AI technology.	3.79	1.23	0.89	0.96	0.96	1.20
	2.	Today, many of the problem-solving activities of this job have been automated or assisted by AI technology.	3.86	1.18	0.90			
	3.	Today, many of the learning activities for this job are automated or assisted by AI technology.	3.74	1.22	0.88			
	4.	Today, much of the reasoning needed for this job is automated or assisted by AI technology.	3.86	1.22	0.90			

Table A1. *Cont.*

Scale	Items	Mean	Standard Deviation	Factor Loading	Cronbach Alpha	Composite Reliability (CR)	VIF
Explicit Knowledge (Choi and Lee, 2003 [83])	1. Knowledge about AI tools (including know-how, technical skills, and problem-solving methods) is well-codified within my company.	3.42	1.52	0.79	0.95	0.95	1.24
	2. Knowledge about AI tools can be easily acquired through formal documents and manuals within my company.	3.33	1.47	0.80			
	3. New AI tools usage tactics should be documented in my company.	3.33	1.54	0.82			
	4. Knowledge about AI tools is shared through codified forms like manuals or documents in my company.	3.41	1.46	0.81			
Tacit Knowledge (Choi and Lee, 2003 [83])	1. My knowledge of AI tools can be easily acquired from experts and co-workers in my company.	3.64	1.36	0.89	0.96	0.97	1.34
	2. It is easy to obtain face-to-face advice on AI tools from experts in my company.	3.67	1.32	0.91			
	3. Informal dialogues and meetings are used for knowledge sharing about AI tools in my company.	3.65	1.34	0.89			
	4. Knowledge about AI tools is acquired through one-to-one mentoring in my company.	3.54	1.34	0.89			
Career Sustainability (Chin et al., 2022 [42])	Resourceful 1. My career enables me to have a good standard of living.	3.67	1.27	0.77	0.95	0.96	DV
	2. My career makes me feel happy because I use my resources well.	3.68	1.17	0.71			
	3. My career makes me feel like I have a bright future.	3.61	1.20	0.77			

Table A1. *Cont.*

Scale	Items			Mean	Standard Deviation	Factor Loading	Cronbach Alpha	Composite Reliability (CR)	VIF
	Flexible	1.	My career allows me to seek new opportunities.	3.60	1.27	0.71			
		2.	My career allows me to learn new things continuously.	3.61	1.20	0.75			
		3.	My career gives me a lot of flexibility.	3.63	1.28	0.77			
	Renewable	1.	My career provides me with opportunities to update my skills.	3.68	1.24	0.79			
		2.	My career gives me the chance to reassess my capabilities.	3.60	1.28	0.73			
		3.	My career enables me to rebrand or reposition myself.	3.60	1.22	0.76			
	Integrative	1.	My career enables me to integrate information obtained from different sources.	3.64	1.32	0.80			
		2.	My career enables me to critically evaluate information obtained from different sources.	3.82	1.17	0.77			
		3.	My career has built my ability to absorb information and knowledge.	3.86	1.10	0.78			

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